SEAT NO:- 31470020

DR. HOMI BHABHA STATE UNIVERSITY

INSTITUTE OF SCIENCE

DEPARTMENT OF BUSINESS AI

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DESIGN THINKING AND INNOVATION - LAB (DTI)

NAME:- SMIT PATIL
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SEMESTER:- III
TEACHER SIGN:- _____

DR. HOMI BHABHA STATE UNIVERSITY

INSTITUTE OF SCIENCE

15, MADAM CAMA ROAD, MUMBAI-400 032

Tel. No. (022) 48904545 Email:hbsu.support@digivarsity.com

DEPARTMENT OF BUSINESS AI CERTIFICATE

This is to certify that **Mr. SMIT PATIL** Exam Seat No./Roll No. 31470020 B.Sc. Business AI (Semester III) has completed the practical work in the subject

Design Thinking and Innovation - Lab (DTI) during the academic year 2026-27 under the guidance of **PROF.** AMINA Ma'am being the partial requirement for the fulfilment of the curriculum of Degree of Bachelor of Science in Business AI at The Institute of Science, Dr. Homi Bhabha State University.

Professor In-Charge **Coordinator**

Date: - ……. **College seal**

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Common Problem Statement for all practicals: Hard to find available study rooms in the library.

Practical 1: Problem Identification and Team Formation

Aim:

To identify and analyse the problem of students finding it difficult to locate available study rooms in the library, and to define a clear problem statement.

Problem Statement:

Students face significant delays and frustration when trying to find an available study room in the library, leading to wasted time and decreased productivity.

Objectives:

1. To identify the key stakeholders involved in library operations.
2. To brainstorm possible root causes for the unavailability of rooms.
3. To define a clear and concise problem statement.

Tools and Materials Required:

Whiteboard, markers, sticky notes, paper, and pens.

Procedure / Methodology:

1. Form a team of 3-4 members.
2. Discuss personal experiences regarding finding study spaces in the library.
3. Identify stakeholders (students, library staff, management).
4. Brainstorm causes using the '5 Whys' technique.
5. Draft the final problem statement.

Observations and Analysis:

Stakeholder Identification

- Primary stakeholders: Undergraduate students and postgraduate researchers, who search for and use the rooms.
- Secondary stakeholders: Library administrative staff, who manage room access, keys and day-to-day complaints.
- Tertiary stakeholders: College management and the IT department, who fund, approve and maintain any solution.

5 Whys Analysis

- Why 1: Why do students spend so long finding a room? Because they cannot tell which rooms are free without checking each one in person.
- Why 2: Why can they not tell from outside? Because there is no real-time display or system showing room status.
- Why 3: Why is there no such system? Because room usage is managed informally, with no booking or tracking process.
- Why 4: Why is it managed informally? Because the library has not digitised its room management.

- Why 5: Why does that make rooms seem unavailable? Because rooms stay blocked by belongings or unused bookings with no way to detect or release them.

Root Causes Identified

- Lack of real-time occupancy tracking.
- Students leaving belongings to 'reserve' rooms (ghost occupancy).
- Uneven distribution of peak hours, with demand clustering before exams and between lectures.

Final Problem Statement

Undergraduate and postgraduate students cannot see which library study rooms are free, so they lose 15-20 minutes walking between floors during limited study time.

Result / Conclusion:

The problem was successfully analysed, stakeholders were identified, and a concrete problem statement was defined to guide further design thinking steps.

Practical 2: User Interviews and Observation

Aim:

To understand students' experiences, challenges, and expectations when searching for available study rooms.

Problem Statement:

Without direct user feedback, it is difficult to grasp the actual pain points students face while searching for library spaces.

Objectives:

1. To prepare a comprehensive interview guide with 10-12 questions.
2. To conduct live interviews with at least three library users.
3. To record and analyse observational data.

Tools and Materials Required:

Interview guide (printed), notebook, voice recorder (optional), pens.

Procedure / Methodology:

1. Draft 10-12 open-ended questions focusing on the user's journey and frustrations.
2. Approach three students in or around the library.
3. Conduct 10-minute interviews and note down their responses.
4. Observe the library environment to see how students search for rooms.

Observations and Analysis:

Interview Guide (12 Questions)

No.	Question
1	How often do you visit the library for group or individual study?
2	How much time do you typically spend looking for a room?
3	What is your biggest frustration when finding a space?
4	How do you currently check if a room is empty?
5	What would make this process easier for you?
6	At what times of day or in which weeks is it hardest to find a room?
7	Have you ever found a room that looked empty but was reserved by belongings?
8	Do you prefer to book a room in advance or find one on arrival? Why?
9	How many people do you usually study with, and what room size do you need?
10	What features (whiteboard, power sockets, quiet zone) matter most to you?
11	What do you do when no room is available?

No.	Question
12	Would you use a mobile app or website to check room status before coming to the library?

Interview Records

Interviewee	Search time	Key insight
1: First-year BSc student	About 15 minutes	Biggest frustration is a room that looks empty but is occupied by bags. Would use an app to check before leaving the hostel.
2: Second-year BSc student	About 18 minutes	Studies in a group of four. Peak difficulty is the week before exams. Wants to pre-book a room with a whiteboard.
3: Postgraduate researcher	About 20 minutes	Needs a quiet room for long sessions. Finds rooms locked or held by belongings. Wants a cancellation or release option.

Observation of the Library Environment

- Students walk along corridors and look through door glass one room at a time.
- Several rooms held only bags, books or jackets with nobody present.
- Students who did not find a room left or studied in crowded open areas.
- There is no sign, display or notice board showing room status.

Key Findings

- Users reported spending 15-20 minutes roaming different floors.
- The biggest frustration is finding a room empty but locked or occupied by bags.
- All three interviewees said they would use a digital tool to check availability.

Result / Conclusion:

Qualitative data was successfully gathered, highlighting that the lack of visibility and 'ghost occupancy' are the primary frustrations for students.

Practical 3: Empathy Map and Persona

Aim:
To synthesize user research into an empathy map and develop a user persona representing the target audience.

Problem Statement:
Raw interview data needs to be structured to build deep empathy for the user's emotional and practical needs.

Objectives:
1. To create an empathy map covering what the user Says, Thinks, Does, and Feels.
2. To develop a realistic user persona based on the target demographic.

Tools and Materials Required:
Large chart paper, sticky notes, markers.

Procedure / Methodology:
1. Review the interview notes from Practical 2.
2. Draw a 4-quadrant Empathy Map (Says, Thinks, Does, Feels).
3. Populate each quadrant with insights.
4. Create a fictional persona summarizing the demographic, goals, and frustrations of the average user.

Observations and Analysis:
Empathy Map: A Student Who Struggles to Find an Available Study Room

SAYS
- 'I wasted 20 minutes looking for a spot again.'
- 'Is anyone actually using this room, or are those just bags?'
- 'I have a group meeting in ten minutes and nowhere to sit.'

THINKS
- 'Why isn't there a system to show which rooms are free?'
- 'If I had known earlier, I could have gone straight to a free room.'
- 'I am losing precious revision time before my exam.'

DOES
- Walks up and down three flights of stairs checking every door.
- Peers through door glass and tries handles of locked rooms.
- Asks other students or library staff whether a room is free.
- Gives up and studies in a noisy open area, or leaves.

FEELS
- Frustrated at the wasted time.
- Anxious before exams.
- Exhausted from walking and searching.

- Annoyed at rooms held by belongings.

User Persona: Studious Sam

Studious Sam is a 19-year-old second-year BSc student who commutes to college and lives on a tight timetable. Sam has about an hour between lectures and needs a quiet room for group study with three classmates. Sam is diligent and tech-comfortable, and uses a smartphone for almost every task, but the library gives no digital way to check room status.

- Demographics: Age 19, BSc student, full-time, daily library user during exam periods.
- Goals: To find a quiet group study room quickly, to start studying within minutes, and to prepare for exams without stress.
- Frustrations: Spends 15-20 minutes walking between floors, finds rooms locked or held by bags, and cannot plan ahead.
- Bio: Sam treats the library as the main place to study but often arrives to find nothing available. Sam would happily use an app that shows free rooms before leaving for the library.

Result / Conclusion:

An empathy map and persona were successfully created, providing a human-centric focus for the ideation phase.

Practical 4: User Journey Map

Aim:

To map the different stages a student experiences while searching for and using a library study room.

Problem Statement:

Identifying the exact touchpoints where the user experiences the most friction requires a structured visual timeline.

Objectives:

1. To outline the chronological steps of the user's journey.
2. To identify pain points and emotions at each stage.
3. To highlight moments of delight or potential improvement.

Tools and Materials Required:

Journey mapping template, sticky notes, markers.

Procedure / Methodology:

1. Define the start and end points of the user journey (from entering the library to settling down).
2. Break the journey into discrete stages.
3. Map the user's actions, thoughts, and emotional state across these stages.
4. Identify specific pain points.

Observations and Analysis:

Journey Stages: Arrival, Floor Navigation, Room Checking, Settling In, Leaving

Stage	Action	Emotion	Pain Point
1. Arrival	Enters the library with a bag and laptop and looks for a notice or display of free rooms.	Hopeful but uncertain (mildly positive)	No signage or screen shows room availability at the entrance.
2. Floor Navigation	Climbs stairs and walks the corridors of each floor.	Impatient and tired (neutral to negative)	Must guess which floor has free rooms, wasting time and energy.
3. Room Checking	Looks through door glass and tries handles, door by door.	Frustrated, annoyed, physically fatigued (lowest point)	Rooms are locked or held by bags, with no way to know if the occupant will return.
4. Settling In	Finds a free room, sets up the laptop and books, and begins studying.	Relieved and satisfied (highest point)	Time lost searching reduces study time; worry that the room may be claimed.
5. Leaving	Packs up and leaves when the session ends.	Neutral, slightly tired	The room's status is not updated, so the next student cannot tell it is free.

Delight Moments

- Finally sitting down and starting the study session.
- Finding a room with a whiteboard and power sockets available.

Pain Point Summary

- The 'Room Checking' phase has the lowest emotional curve, categorized by physical fatigue and annoyance.
- 'Floor Navigation' and 'Room Checking' together account for most of the 15-20 minutes lost.

Result / Conclusion:

The user journey map revealed that the 'Floor Navigation and Room Checking' stages are the critical bottlenecks requiring a design intervention.

Practical 5: Define - POV and How Might We Statements

Aim:

To formulate a Point of View (POV) statement and How Might We (HMW) statements based on research findings.

Problem Statement:

The broad problem needs to be narrowed down into actionable design challenges.

Objectives:

1. To write a strong POV statement based on user needs.
2. To generate at least three HMW statements.
3. To select the best HMW statement for ideation.

Tools and Materials Required:

Paper, pens, whiteboard.

Procedure / Methodology:

1. Use the format: [User] needs to [Need] because [Insight].
2. Brainstorm HMW questions that reframe the POV into an opportunity.
3. Vote on the most impactful HMW statement.

Observations and Analysis:

Point of View (POV) Statement

A time-pressed BSc student needs a quick, reliable way to secure a study room because randomly wandering the library wastes valuable study time and causes pre-exam anxiety.

How Might We (HMW) Statements

No.	HMW Statement	Evaluation
1	How might we make room availability visible before students even enter the library?	Addresses the biggest finding (lack of visibility) and the lowest point of the journey map.
2	How might we prevent students from leaving belongings to 'block' rooms?	Important, but depends on policy and enforcement.
3	How might we streamline the room booking process?	Useful, but only helps once students can see what is free.

Selected Statement

How might we make real-time room availability visible to students?

Result / Conclusion:

A clear POV and actionable HMW statements were defined, setting a focused direction for generating solutions.

Practical 6: Ideation Workshop

Aim:

To generate innovative solutions to help students locate available library study rooms and select the most feasible one.

Problem Statement:

Without structured brainstorming, solutions may be too conventional or technically unfeasible.

Objectives:

1. To generate at least five diverse ideas.
2. To evaluate ideas using an Impact-Effort matrix.
3. To select the best solution.

Tools and Materials Required:

Sticky notes, markers, Impact-Effort matrix chart.

Procedure / Methodology:

1. Conduct a 15-minute open brainstorming session.
2. Write one idea per sticky note.
3. Draw an Impact (Low-High) vs. Effort (Low-High) matrix.
4. Place ideas on the matrix and choose a high-impact, low/medium-effort solution.

Observations and Analysis:

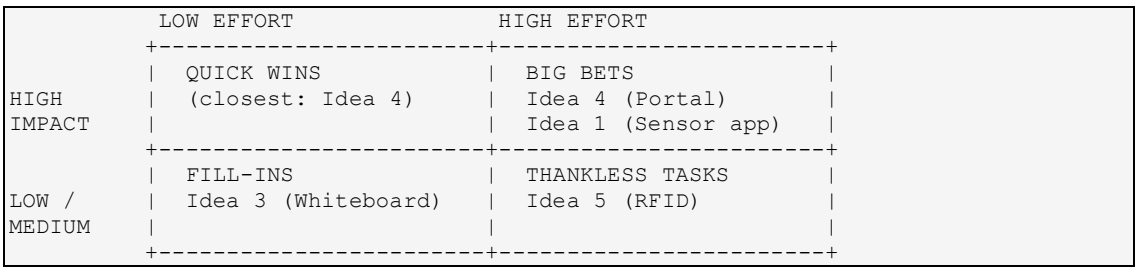
Ideas Generated

- Idea 1: Real-time mobile app linked to motion sensors.
- Idea 2: Digital traffic-light display outside rooms.
- Idea 3: Library staff manually updating a whiteboard.
- Idea 4: Pre-booking web portal.
- Idea 5: RFID tags on student IDs to track occupancy.

Impact-Effort Evaluation

Idea	Impact	Effort	Quadrant
1. Sensor-linked mobile app	High	High	Major Project
2. Digital traffic-light display	Medium	Medium	Possible Quick Win
3. Manual whiteboard	Low	Low	Fill-in
4. Pre-booking web portal	High	Medium	Best Choice
5. RFID student ID tracking	Medium	High	Reconsider

Impact-Effort Matrix



Selection

The 'Pre-booking web portal and digital availability display' (combining Ideas 4 and 2) was selected (High Impact, Medium Effort). It directly addresses the selected HMW statement and can be built without specialised hardware.

Result / Conclusion:

Five ideas were generated and evaluated. A digital booking portal with real-time display was chosen as the most viable solution.

Practical 7: Storyboarding the Solution

Aim:

To draw a six-frame storyboard illustrating how a student experiences the proposed solution.

Problem Statement:

Stakeholders need to visualize how the proposed solution fits into the user's daily life.

Objectives:

1. To illustrate the 'before' state of the user.
2. To show the interaction with the new solution.
3. To demonstrate the 'after' state and benefits.

Tools and Materials Required:

Storyboard template (6 frames), pencils, erasers.

Procedure / Methodology:

1. Sketch Frame 1: User facing the problem (frustration).
2. Sketch Frame 2: User discovering/accessing the solution.
3. Sketch Frame 3: User interacting with the main feature (e.g., booking).
4. Sketch Frame 4: User executing the physical action (walking to the room).
5. Sketch Frame 5: User experiencing the benefit (studying peacefully).
6. Sketch Frame 6: The conclusion (system updates availability).

Observations and Analysis:

Six-Frame Storyboard Script

Frame	Visual Description	Action / Dialogue
1 [Wide Shot: Before]	A stressed student stands at the library entrance holding a laptop and notebooks, looking up at the staircase, with a clock showing 2:00 PM.	Student sighs: "Not again. Which floor is free today?"
2 [Medium Shot: Discovery]	The student takes out a phone and opens the 'Library Space' app, whose home screen shows a floor map.	Student: "Maybe I can check before I start walking."
3 [Close-up: Booking]	The phone shows Room 204 in green (Available) and other rooms in red. The student's thumb taps the 'Book' button.	A confirmation appears. Student: "Room 204 is free, and it is mine for two hours."
4 [Tracking Shot: Walking]	The student walks directly up the stairs to the second floor, passing closed doors without checking them.	Student (smiling): "No more door-hopping."

Frame	Visual Description	Action / Dialogue
5 [Medium Shot: Benefit]	The student sits in Room 204 with books and laptop open, a friend beside them, and the clock now reads 2:10 PM.	Student: "Ten minutes, not twenty. I can actually start revising."
6 [Wide Shot: Conclusion]	The student leaves at the end of the session, and the app on the phone shows Room 204 turning green again.	App notification: "Room 204 is now available." The next student sees it and books it.

Result / Conclusion:

The storyboard successfully visualized the seamless integration of the booking app into the student's routine, highlighting time saved.

Practical 8: Low-Fidelity Prototype

Aim:

To develop a paper prototype or wireframe for the proposed library study-room availability system.

Problem Statement:

Moving straight to coding without testing the user interface can lead to poor user experience.

Objectives:

1. To design a minimum of three core screens.
2. To include room availability status, room details, and confirmation.
3. To ensure logical navigation between screens.

Tools and Materials Required:

A4 paper, pencils, rulers, wireframing templates.

Procedure / Methodology:

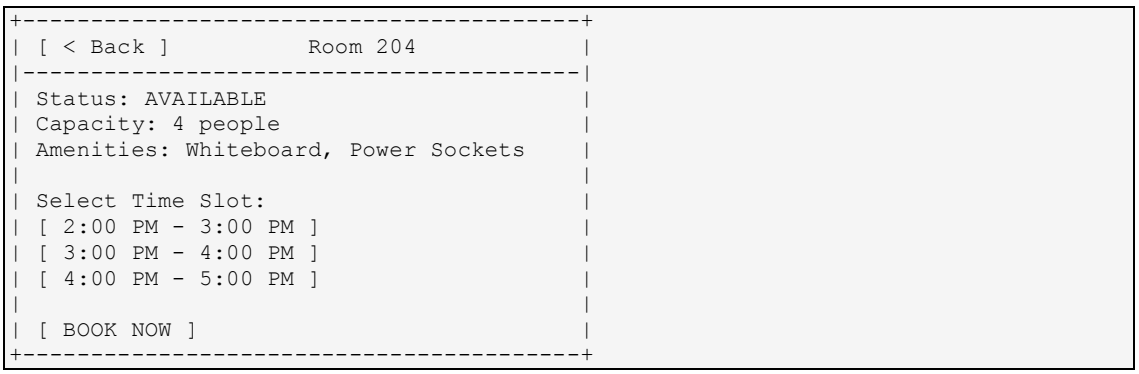
1. Sketch the mobile device outline.
2. Design Screen 1: Dashboard with a floor map and color-coded availability.
3. Design Screen 2: Room Details page showing capacity, whiteboard availability, and time slots.
4. Design Screen 3: Booking Confirmation screen with a QR code or summary.

Observations and Analysis:

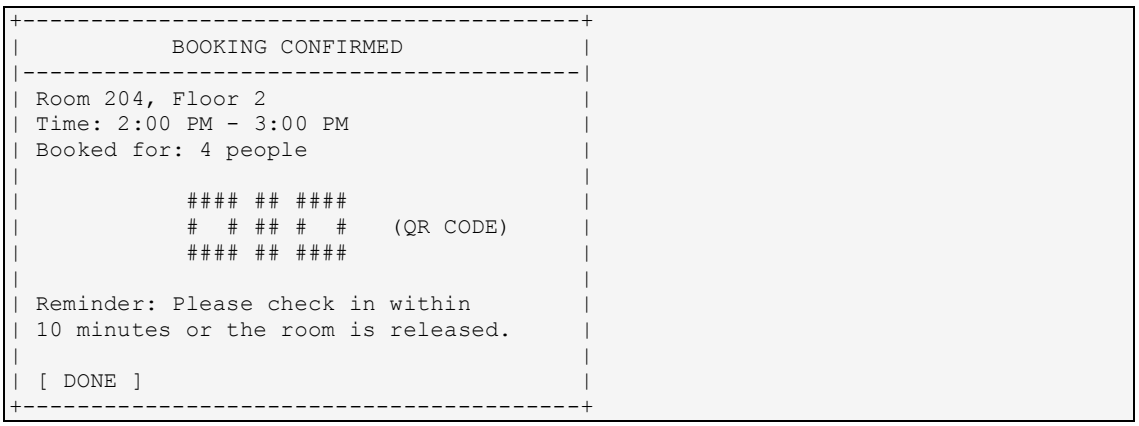
Screen 1: Dashboard (Room Availability)

```
+----------------------------------------+
| [Library Logo]    Floor: 2  [v]        |
|----------------------------------------|
| Floor 2 Map                            |
|                                        |
| Room 201: [ GREEN - AVAILABLE ]        |
| Room 202: [ RED - OCCUPIED ]           |
| Room 203: [ RED - OCCUPIED ]           |
| Room 204: [ GREEN - AVAILABLE ]        |
|                                        |
| Legend:  GREEN = Free   RED = Occupied |
|----------------------------------------|
| [ Floor 1 ]  [ Floor 2 ]  [ Floor 3 ]  |
+----------------------------------------+
```

Screen 2: Room Details



Screen 3: Booking Confirmation



Navigation Flow

- Screen 1 (tap a green room) leads to Screen 2.
- Screen 2 (tap Book Now) leads to Screen 3.
- Screen 3 (tap Done) returns to Screen 1.
- The Back button on Screen 2 returns to Screen 1.

Design Notes

- Screen 1 features a clear toggle between floors and red/green room indicators.
- Screen 2 includes a 'Book Now' button and amenities list.
- Screen 3 provides a simple success message and time limit reminder.

Result / Conclusion:

A tangible, low-fidelity paper prototype was created, ready for initial usability testing.

Practical 9: Usability Testing and Iteration

Aim:
To test the prototype with users, collect feedback, identify problems, and iterate on the design.

Problem Statement:
The initial prototype may contain navigation flaws or missing features that only real users can identify.

Objectives:
1. To prepare specific usability tasks.
2. To test with at least three students.
3. To update the prototype based on feedback.

Tools and Materials Required:
Paper prototypes from Practical 8, notebook, pens.

Procedure / Methodology:
1. Define tasks (e.g., 'Book a room for 4 people at 2 PM').
2. Have users interact with the paper screens (Wizard of Oz testing).
3. Note down where they hesitate or get confused.
4. Revise the prototype sketches.

Observations and Analysis:

Usability Tasks
- Task 1: Find an available room on Floor 2.
- Task 2: Book Room 301 for 4 people at 2 PM.
- Task 3: Cancel the booking because your plans have changed.

Test Results (3 Participants)

Task	Student 1	Student 2	Student 3	Result
Task 1: Find a room	Completed	Completed	Completed	3/3 success
Task 2: Book Room 301	Completed quickly	Completed quickly	Completed quickly	3/3 success
Task 3: Cancel booking	Could not find option	Could not find option	Could not find option	0/3 success

User Feedback
- Users found the booking fast and the red/green colours easy to understand.
- Users noted there was no way to cancel a booking if their plans changed.
- Users could not tell how much of their booked time remained once the session began.

Usability Problems Identified
- Problem 1 (critical): No cancellation option, so unused rooms stay blocked.

- Problem 2 (moderate): No time-remaining indicator after booking.

Iteration

A 'Cancel Booking' button and a 'Time Remaining' timer were added to the dashboard.

Revised Dashboard (after iteration)

```
+----------------------------------------+
| [Library Logo]    Floor: 2  [v]        |
|----------------------------------------|
| Your Booking: Room 204                 |
| Time Remaining: 00:42:15               |
| [ CANCEL BOOKING ]                     |
|----------------------------------------|
| Room 201: [ GREEN - AVAILABLE ]        |
| Room 202: [ RED - OCCUPIED ]           |
| Room 203: [ RED - OCCUPIED ]           |
+----------------------------------------+
```

Result / Conclusion:

Testing revealed a critical missing feature (cancellation). The prototype was successfully iterated to improve user control.

Practical 10: Final Pitch - Business Model and MVP

Aim:

To prepare a final pitch, including a Business Model Canvas (BMC) and a description of the Minimum Viable Product (MVP).

Problem Statement:

The solution needs a sustainable implementation strategy and a pitch to convince college management.

Objectives:

1. To prepare a Business Model Canvas.
2. To define the Minimum Viable Product.
3. To deliver a structured pitch explaining features, benefits, and feasibility.

Tools and Materials Required:

Presentation software (PPT), BMC template, projector.

Procedure / Methodology:

1. Fill out the 9 blocks of the BMC (Value proposition, Customer segments, etc.).
2. Define what features the MVP will have for initial launch.
3. Structure a 5-minute pitch covering the problem, solution, and roadmap.

Observations and Analysis:

Solution Summary

A study-room availability and booking system ('Library Space') that shows live room status by floor, lets students book and cancel rooms, and releases unused rooms automatically.

Key Features

- Colour-coded floor map (green available, red occupied).
- Room details with capacity, amenities and time slots.
- Booking confirmation with QR code and check-in reminder.
- Cancel Booking button and Time Remaining timer.

Benefits

- Saves the 15-20 minutes students currently spend searching.
- Reduces pre-exam anxiety and frustration.
- Reduces ghost occupancy and gives staff clearer room-usage data.

Feasibility

- Technical: Needs only a responsive website and a simple database, with no special hardware for the first version.
- Operational: Library staff can update the room status manually at first.
- Financial: Low development and hosting cost, funded within the university's IT budget.

Business Model Canvas

<table>
<tr>
<td>Key Partners
• College management and IT department
• Library administration
• Web hosting provider</td>
<td>Key Activities
• Developing and maintaining the booking system
• Updating room availability
• Collecting student feedback
Key Resources
• Development team and library staff
• Servers and hosting
• Room and booking database</td>
<td>Value Propositions
• Save time and reduce anxiety
• See room availability before reaching the library
• Fair, transparent booking with cancellation</td>
<td>Customer Relationships
• Self-service through website or app
• Staff support at the library desk
• Feedback form
Channels
• Responsive website (MVP)
• Mobile app (later phase)
• Notice at library entrance with web link</td>
<td>Customer Segments
• University students (undergraduate and postgraduate)
• Library staff as secondary users</td>
</tr>
<tr>
<td colspan="3">Cost Structure
• App development
• Server hosting
• Maintenance and staff time</td>
<td colspan="2">Revenue Streams
• Funded by the university's IT budget (no fee charged to students)</td>
</tr>
</table>

Minimum Viable Product (MVP)

A simple responsive website (no native app yet) where library staff manually update room availability every 30 minutes, allowing students to view status without booking.

MVP Testing Plan

- Launch on one floor for two weeks.
- Measure the average time students take to find a room against the current 15-20 minutes.
- Collect feedback through a short survey, then add booking and cancellation in the next phase.

5-Minute Pitch Structure

Segment	Time
Problem and user pain	1 minute
Solution and key features	2 minutes
Business model and MVP	1 minute
Roadmap and request for approval	1 minute

Result / Conclusion:

A comprehensive business model and a clear, low-cost MVP strategy were formulated, preparing the team for real-world implementation.